UNIVERSITY OF OULU · BOTANICAL GARDEN · MAY 2026

# PhytoSense

Continuous Environmental Monitoring

*Romeo & Julia hold 1,200 irreplaceable exotic species.*
*A Finnish winter night can silently stress them all.*
*Nobody is currently listening.*

# The Problem

*Manual floor rounds happen twice a day, at best.*

The system catches the 3am humidity accumulation in the tropical corner that nobody would find until morning.

### Fungal risk

Botrytis & pathogens establish overnight in high humidity

### Watering gaps

Overwatered succulents, parched tropicals — between rounds

### Winter stress

Glass-wall cold spots damage specimens silently in Finnish winter

# What PhytoSense Does

*A wireless mesh of sensor nodes installed across one greenhouse zone.*

Each node carries three sensors that report continuously to a central hub. Staff receive real-time alerts. No cloud services. No ongoing costs. Self-contained on the garden's local network.

**Temperature**

**Humidity**

**Soil Moisture**

EUR 220

Total MVP hardware

1,200

Exotic species monitored

24 / 7

Continuous

EUR 0

Software cost (open source)

# Problem 1 — Fungal Risk Conditions

*Botrytis*
*& Mould*

### What the system detects

Sustained humidity + temperature combinations that favour pathogen development

### When it matters most

Overnight when no one is on the floor — humidity accumulates for hours undetected until morning rounds

### What staff receive

An alert the moment conditions enter the risk window — not after damage appears

# Problem 2 — Watering Problems

*Soil Moisture*

## The challenge

A desert succulent and a tropical orchid have entirely different moisture needs. One threshold fits neither.

### Overwatered

Root rot, fungal growth, soil compaction — the most common cause of exotic specimen loss in mixed greenhouses

### Underwatered

Heat stress, leaf drop, irreversible damage to decades-old specimens before any visual sign appears

*PhytoSense sets per-zone thresholds agreed directly with horticulture staff — not a single global setting.*

# Problem 3 — Environmental Suitability

*Finnish Winter*

## Conditions the system watches continuously

### Cold spots near the glass walls

Romeo's tropical corner can fall below safe limits on Finnish winter nights — long before a staff member walks the floor.

### Humidity collapse

Heating systems dry the air. The tropical zone can lose safe humidity levels within hours during cold snaps.

### Summer heat accumulation

Glass greenhouses trap heat. Sensitive specimens — orchids, carnivorous plants, decades-old rarities — need early warning.

# How It Works

01

**Sensor Nodes**

Four ESP32 nodes, each with soil moisture, humidity and temperature sensors. Battery powered, wireless, no cabling required.

02

**WiFi Mesh**

Nodes communicate continuously over the greenhouse WiFi network to a central Raspberry Pi hub on the garden's local network.

03

**Staff Alerts**

Home Assistant on the Pi logs all data and fires real-time alerts to horticulture staff when any threshold is crossed.

04

**Two Dashboards**

A staff-only view with zone status and alert history, plus a visitor-facing exhibit showing live plant environment conditions.

# Two Dashboards, One Data Source

## Staff Dashboard

Colour-coded zone status — green, amber, red

- Real-time threshold alerts to staff devices
- Full data history and overnight logs
- Thresholds configured with horticulture staff at installation
- Internal only — no public access to operational data

## Visitor Exhibit

- Live environmental conditions framed as the plant experience
- "Tropical corner: warm and humid — comfortable"
- No operational alerts — no data that reflects on maintenance
- Invites visitors into the science happening in real time
- Interactive exhibit at zero additional hardware cost

# Cost & Scale

# EUR 220

Total Phase 1 MVP hardware cost

*Commercial systems with equivalent capability: several thousand euros*

## Phase 1 Components

| | |
|---|---|
| ESP32 microcontrollers (x5) | ~EUR 40 |
| Raspberry Pi 4 hub (2GB) | ~EUR 55 |
| SHT31-D humidity + temp sensors (x4) | ~EUR 18 |
| Soil moisture sensors (x8) | ~EUR 12 |
| 18650 batteries & charge modules (x4) | ~EUR 22 |
| Enclosures, PCB, cables | ~EUR 47 |

**Phased rollout — committee asked to fund Phase 1 only**

### Phase 1

**4 nodes · one zone**

This proposal. Staff alerts live from installation.

### Phase 2

**10–12 nodes · Romeo**

Full tropical and subtropical coverage.
First seasonal dataset.

### Phase 3

**16–20 nodes · Both**

Romeo and Julia covered.
Full public exhibit.
Solar-powered nodes.

*Thank you!*

*Questions?*